INVENTION EXHIBITION

Smart Bicycle Tail Light System

智慧自行車尾燈系統

WIRELESS CONTROL
MOTION SENSING
LOCAL ALERTS

Inventor / School / Advisor / Exhibition: to be confirmed



Physical demonstration model 實體展示模型

THE DESIGN GOAL

Help people behind a bicycle recognize turns, braking and emergency warnings.

讓後方用路人更容易辨識轉彎、煞車及緊急警示。



Rear light installed on a bicycle
實車安裝

INPUT
micro:bit V2
Buttons, touch logo, acceleration

→

BLE LINK
Native services
Button, Accelerometer, Event

→

PROCESS
XIAO ESP32-C6
Calibration, filtering, state logic

→

OUTPUT
8 LEDs + buzzer
LED: D1　Buzzer: D3
Brightness capped in software

Native micro:bit BLE services, without UART. 400 mAh lithium battery. Set the forward sensor axis and sign.

SIX CORE FEATURES 六大功能

01 Riding and parking

Soft red with two green center LEDs while riding. Steady red after a configured period at rest.

巡航為紅燈加中央兩顆綠燈；靜止後切換全紅。

02 Braking alerts

Moving averages and slope tracking filter acceleration. A braking trigger starts red flashes and a buzzer.

開機校正、移動平均與坡度追蹤；減少誤觸的效果待測。

03 Turn signals

A: left. B: right. Four LEDs on the turning side show an amber sequence with sound. The other side keeps its lighting. Auto-cancel or timeout ends the signal.

同側四顆燈顯示方向；另一側維持照明，可自動取消。

04 Hazard lights

Short-press A+B to turn synchronized amber warning lights on or off.

短按 A+B，開啟或解除雙黃警示。

05 SOS warning

Alternating red and white light with an alarm. Hold A+B continuously for 3 seconds to clear SOS.

紅白交替閃光與警報音；連續長按 A+B 三秒解除。

06 Power-saving standby

After prolonged inactivity, the lights turn off. Movement restores them. The whole system stays powered.

長時間不動就關燈，移動後恢復；並非整套斷電。

OPERATING MODES 兩種模式

1 Demonstration mode

Upright calibration, sensitive braking and a sustained-tilt SOS trigger for the model.

展示模式：先扶正校正，以短距離動作與持續傾倒展示。

2 Riding mode

Different braking, impact and tilt parameters aim to reduce unwanted triggers. Validation is pending.

騎乘模式：調整判斷參數；降低誤觸的效果仍待驗證。

Switch with the gold logo or the phone webpage.

OFFLINE PHONE SETTINGS 手機離線設定

About 49 parameters: sensitivity, timing, filtering, tilt angle, lights, sound and standby.

Local Wi-Fi webpage. No app or internet required.

約 49 項可調參數，手機連接本地 Wi-Fi 網頁即可設定。

TESTING AND RECORDS 測試與紀錄

1-100 rounds, default 30

Preparation and observation periods log triggers and acceleration. Manually confirm actual braking.

Exclude unperformed or unconfirmed rounds from the braking success-rate denominator.

實際煞車與否需人工確認，未做或未確認者不列入分母。

Verify actual light output by eye or video.

EVIDENCE AND CURRENT LIMITS 證據與目前限制

Braking detection rate = triggered confirmed braking rounds / all confirmed braking rounds x 100%.

CSV exports: trial results, acceleration, settings and summary. Offline reports compare outcomes and thresholds.

Compilation and computer-side checks completed. Detection rate, false-trigger rate and runtime remain unmeasured.

SOS warns nearby people only. No GPS, remote notification or battery / charge-status sensing. No road-use certification.

成功率、誤觸率及續航待測；SOS 只在現場警示，無定位與遠端通報，尚未取得道路使用認證。